III B.TECH – II SEMESTER

Sustainability In Engineering Practices

Open Elective-II | Code: 23A01606b | JNTUA R23 · B.Tech CSE

Course Code	Sustainability in Engineering practices (OE – II)	L	T	P	C
23A01606b		3	0	0	3
Course Objectives: **The objectives of this course are to make the student :** To understand the fundamentals of sustainability, the carbon cycle, and the environmental impact of construction materials. To analyze sustainable construction materials, their durability, and life cycle assessment. To apply energy calculations in construction materials and assess their embodied energy. To evaluate green building standards, energy codes, and performance ratings. To assess the environmental effects of energy use, climate change, and global warming.					
Course Outcomes: **After successful completion of this course, students will be able to:** Understand the fundamentals of sustainability, the carbon cycle, and the environmental impact of construction materials. Analyze sustainable construction materials, their durability, and life cycle assessment. Apply energy calculations in construction materials and assess their embodied energy. Evaluate green building standards, energy codes, and performance ratings. Assess the environmental effects of energy use, climate change, and global warming.					
CO – PO Articulation					

CO – PO Articulation Matrix **Course Outcomes PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO 10 PO 11 PO 12 PSO1 PSO2 CO -1** 3 - - - - 2 3 2 - - - - 3 3 **CO -2** - 3 - - 2 - 3 - - - - 2 3 3 **CO -3** - - 3 3 3 - 2 - - 2 - - 3 3 **CO -4** - - 3 3 3 - 3 2 - - - - 3 3 **CO -5** - - - - - 3 3 3 - - - - - 3					
UNIT – I					
INTRODUCTION Introduction and Definition of Sustainability - Carbon Cycle - Role of Construction Material: Concrete and Steel, Etc. - CO2Contribution From Cement and Other Construction Materials.					
UNIT – II					
MATERIALS USED in SUSTAINABLE CONSTRUCTION Construction Materials and Indoor Air Quality - No/Low Cement Concrete - Recycled and Manufactured Aggregate - Role of QC and Durability - Life Cycle and Sustainability.					
UNIT – III					
ENERGY CALCULATIONS Components of Embodied Energy - Calculation of Embodied Energy for Construction Materials - Energy Concept and Primary Energy - Embodied Energy Via-A-Vis Operational Energy in Conditioned Building - Life Cycle Energy Use					
UNIT – IV					
GREEN BUILDINGS Control of Energy Use in Building - ECBC Code, Codes in Neighboring Tropical Countries - OTTV Concepts and Calculations – Features of LEED and TERI – Griha Ratings - Role of Insulation and Thermal Properties of Construction Materials - Influence of Moisture Content and Modeling					

- Performance Ratings of Green Buildings - Zero Energy Building					
UNIT – V					
ENVIRONMENTAL EFFECTS Non-Renewable Sources of Energy and Environmental Impact– Energy Norm, Coal, Oil, Natural Gas - Nuclear Energy - Global Temperature, Green House Effects, Global Warming - Acid Rain: Causes, Effects and Control Methods - Regional Impacts of Temperature Change.					
TEXT BOOKS:					
Charles J Kibert, Sustainable Construction: Green Building Design & Delivery, 4th Edition , Wiley Publishers 2016. Steve Goodhew, Sustainable Construction Process, Wiley Blackwell,UK, 2016.					
REFRENCE BOOKS:					
Craig A. Langston & Grace K.C. Ding, Sustainable Practicesin the Built Environment, Butterworth Heinemann Publishers, 2011. William P Spence, Construction Materials, Methods & Techniques (3e), Yesdee Publication Pvt. Ltd, 2012.					
Online Learning Resources:					
https://archive.nptel.ac.in/courses/105/105/105105157/					